

Day 03 Assignment - Network Interfaces and Cables

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THEORY

1. What is a Network Interface Card?

- It is responsible for providing network connectivity to machines. For example, Ethernet NIC.

2. What is Ethernet?

- It is a wired-LAN technology available in different speeds like 10 Mbps, 100 Mbps, 1 Gbps, 10 Gbps, etc.

3. What is RJ-45?

- It is an 8-position modular connector used on Ethernet cables which are found on routers, switches, PCs, etc.

4. What does UTP stand for?

- UTP stands for Unshielded Twisted Pairs.

5. Name three common Ethernet cable categories.

- Categories: Cat5e, Cat6, Cat6a

6. What is a straight-through cable?

- It is used to connect different device types like PC to Switch, Switch to Router, etc.

7. What is a crossover cable?

- It is used to connect similar device types like Switch to Switch, Router to Router, etc.

8. Explain Auto MDI-X.

- When enabled, it automatically detects the required transmit and receive wiring, so the type of cable does not matter.

9. Difference between copper and fiber-optic networking?

- Copper cables use electrical pulses transmitted through metal wiring. It is affordable and easy to install but is limited by distance and electromagnetic interference.
- Fiber-optic cables use light pulses transmitted through tiny strands of glass or plastic wiring. It can be used across long distances and is not affected by electromagnetic interference.

10. What is multimode fiber?

- It is a shorter distance fiber cable commonly used inside buildings, data centres, etc.

11. What is single-mode fiber?

- It is a longer distance fiber used by Telecom, Internet Service Provider, etc.

12. Two advantages of fiber-optic networking.

- It can be used across longer distances.
- It is immune to electromagnetic inference.

13. Why is understanding network interfaces important for cybersecurity?

- Network interfaces are used in various tools like Wireshark, Nmap, etc which are used by cybersecurity professionals.

PRACTICAL IDENTIFICATION

14. Identify your active network adapter.

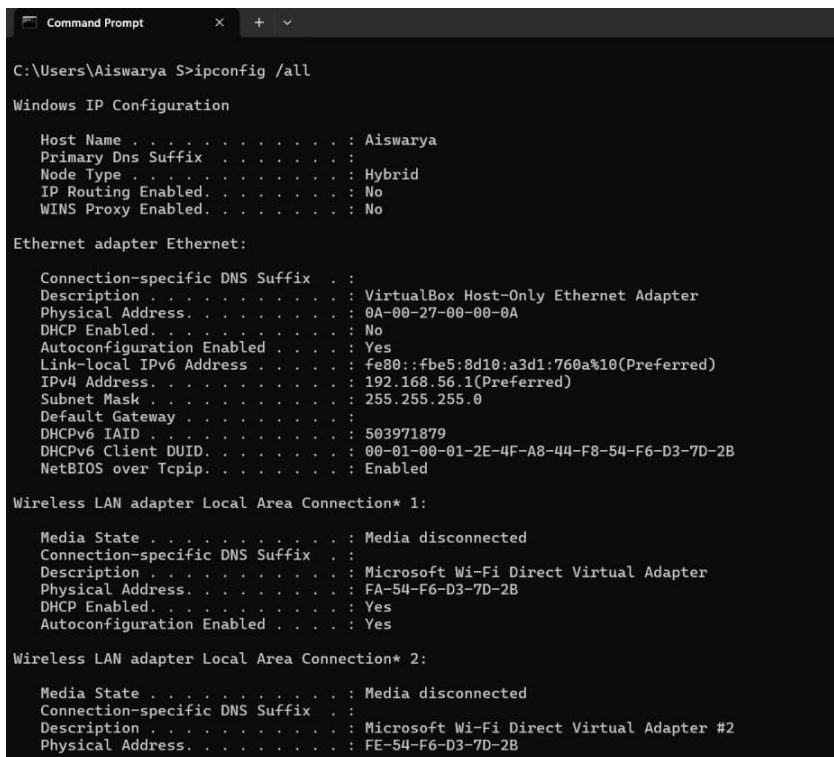
- Wireless LAN adapter Wi-Fi

15. Write your IPv4 address, default gateway, and network adapter name.

- IPv4 Address: 192.168.1.37
- Default Gateway: 192.168.1.1
- Network Adapter Name: Wireless LAN adapter Wi-Fi

SCREENSHOT

- Ipconfig /all



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Command Prompt
C:\Users\Aiswarya>ipconfig /all

Windows IP Configuration

Host Name . . . . . : Aiswarya
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-0A
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::f8e5:8d10:a3d1:760a%10(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 503971879
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-4F-A8-44-F8-54-F6-D3-7D-2B
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : FA-54-F6-D3-7D-2B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : FE-54-F6-D3-7D-2B
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Command Prompt
Physical Address. . . . . : FE-54-F6-D3-7D-2B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : bbrouter
Description . . . . . : Realtek 8821CE Wireless LAN 802.11ac PCI-E NIC
Physical Address. . . . . : F8-54-F6-D3-7D-2B
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . : fe80::c2ef:57b6:d5f2:dda7%11(Preferred)
IPv4 Address. . . . . : 192.168.1.37(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 09 September 2026 14:12:23
Lease Expires . . . . . : 11 September 2026 09:11:30
Default Gateway . . . . . : fe80::1%11
                          192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 116937974
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-4F-A8-44-F8-54-F6-D3-7D-2B
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : F8-54-F6-D3-7D-2A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

Connection-specific DNS Suffix . :
Description . . . . . : Hyper-V Virtual Ethernet Adapter
Physical Address. . . . . : 00-15-5D-EA-57-2A
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . : fe80::5c2d:ea81:ffe4:d02f%36(Preferred)
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Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . : fe80::c2ef:57b6:d5f2:dda7%11(Preferred)
IPv4 Address. . . . . : 192.168.1.37(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 09 September 2026 14:12:23
Lease Expires . . . . . : 11 September 2026 09:11:30
Default Gateway . . . . . : fe80::1%11
                          192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DHCPv6 IAID . . . . . : 116937974
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-4F-A8-44-F8-54-F6-D3-7D-2B
DNS Servers . . . . . : 192.168.1.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : F8-54-F6-D3-7D-2A
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter vEthernet (WSL (Hyper-V firewall)):

Connection-specific DNS Suffix . :
Description . . . . . : Hyper-V Virtual Ethernet Adapter
Physical Address. . . . . : 00-15-5D-EA-57-2A
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-Local IPv6 Address . . . . : fe80::5c2d:ea81:ffe4:d02f%36(Preferred)
IPv4 Address. . . . . : 172.25.176.1(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 603985245
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-4F-A8-44-F8-54-F6-D3-7D-2B
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\Aiswarya S>|
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